CRICKETLENS

Stage 2 / Session 01 - Initial Exploratory Data Analysis

Men's IPL | Source: Cricsheet JSON | Prepared: 18 October 2026

Development scope	Statistics and analysis	Status
2008-2025 (1,169 matches)	First-innings seasonal trends, missingness	Measured
2018-2025 (533 first innings)	Score distribution, phases, players, venues, correlations	Measured
2022-2025 (current franchises)	Decided-match team win rates	Measured
2026	Reserved as ML evaluation holdout; not analyzed here	Protected

Research objectives

Understand scoring distributions and variation; identify patterns by season, innings phase, team, venue and batter; examine associations without mistaking them for causation or forecasts. The analysis supports the eventual IPL Overview, Team Analysis, Player Explorer and Statistics & EDA Lab dashboard tabs.

Verified highlights

Measure	Result	Important context
Recent first-innings mean	175.80 runs	533 first innings in 2018-2025
Recent first-innings median	175 runs	Q1=155, Q3=199, IQR=44
Death phase run rate	10.73 per six legal balls	Only innings that reach death overs
Recorded batter 30+ rate	27.03%	8,123 batting appearances, not all team selections
Next-over wicket frequency	27.85%	19,290 candidate eligible windows (2018-2025)

Methodological boundaries

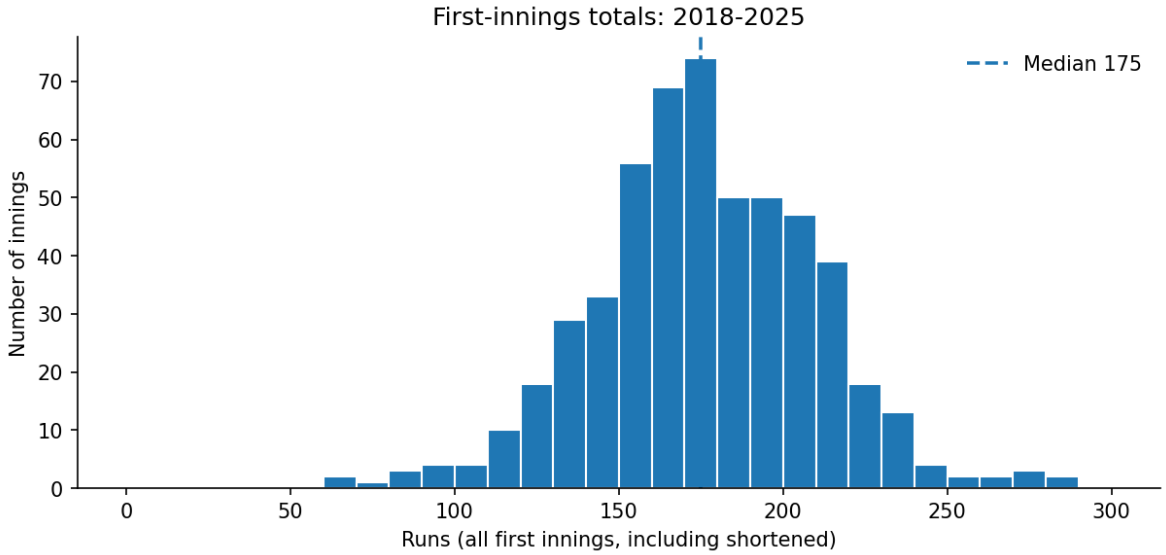
All figures are historical observational summaries. Shortened or all-out innings are retained in general first-innings comparisons; venue and team compositions change across seasons. No actual 2026 model outcomes were used to design these results. Final ML training and testing remain Stage 3 work.

Status: Stage 2 has started successfully, but is not yet signed off as complete.

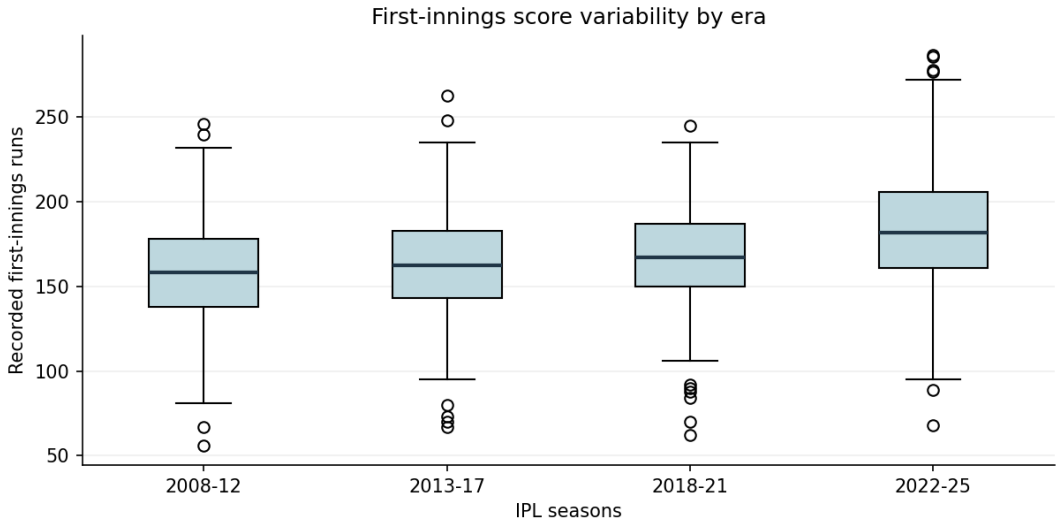
01 Distribution and variability

Sample size	Mean	Median	Sample SD	IQR	Median abs. dev.
533	175.80	175	33.93	44	22

Histogram: first-innings totals, 2018-2025



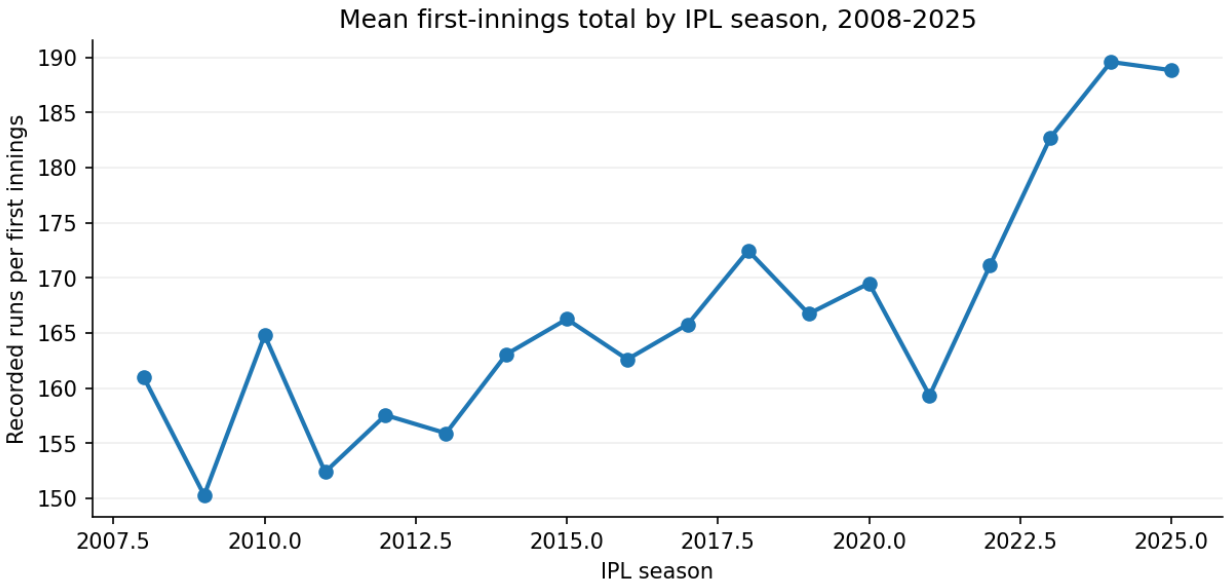
Boxplots: changes in scoring distribution across eras



Interpretation: The recent median is 175; the middle 50% of totals lies from 155 to 199. Tukey fences below 89 or above 265 flag 11 unusual 2018-2025 totals. Outliers are unusual match performances, not automatically corrupted data.

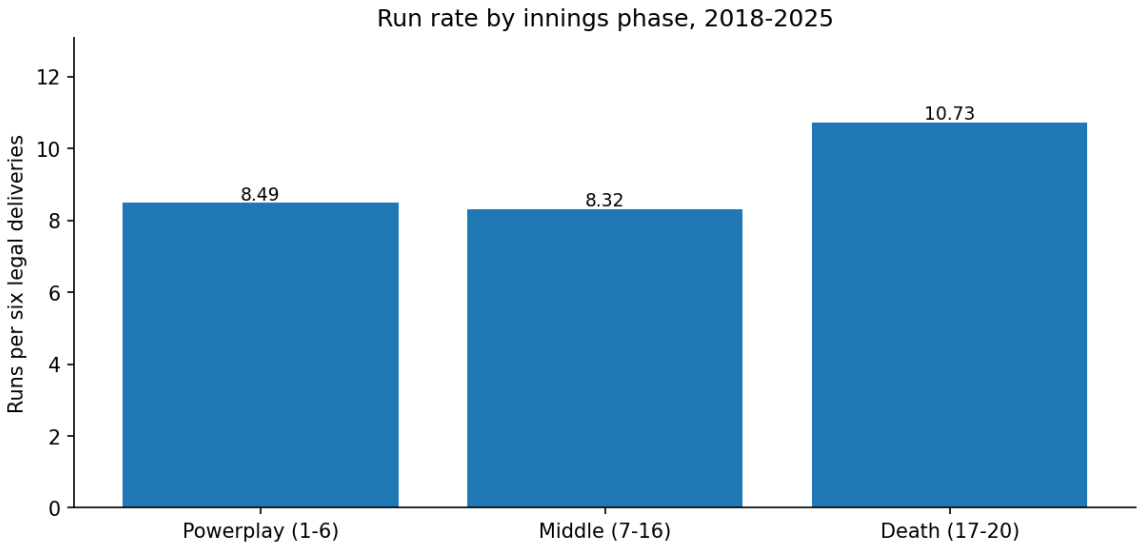
02 Scoring patterns across time and innings

Mean recorded first-innings runs by season (2008-2025)



An unweighted regression line across 18 yearly mean scores has slope +1.71 runs/year (R^2=0.663). This is a descriptive, noncausal association with year; conditions, teams and venues change.

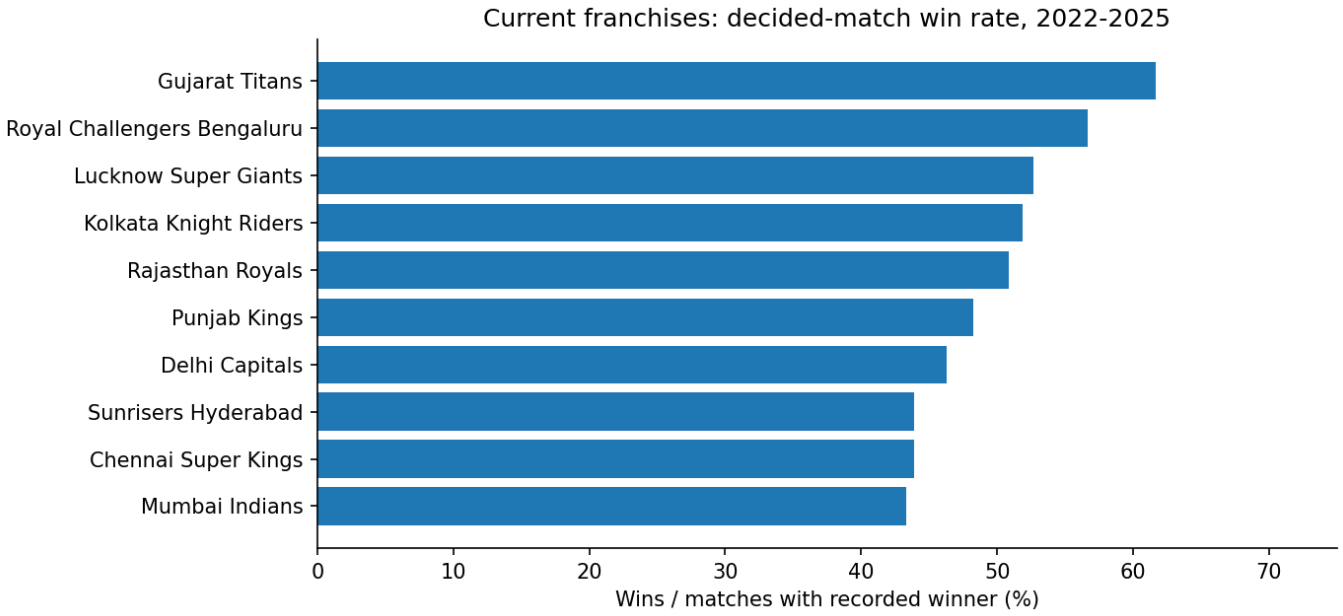
Run rate by Powerplay, middle and death phases, 2018-2025



Phase run rate is total runs divided by the number of legal deliveries, multiplied by six. Wides and no-balls contribute runs but do not count as legal deliveries. Late-innings survivors and bowling usage differ from earlier phases, so these are not controlled comparisons.

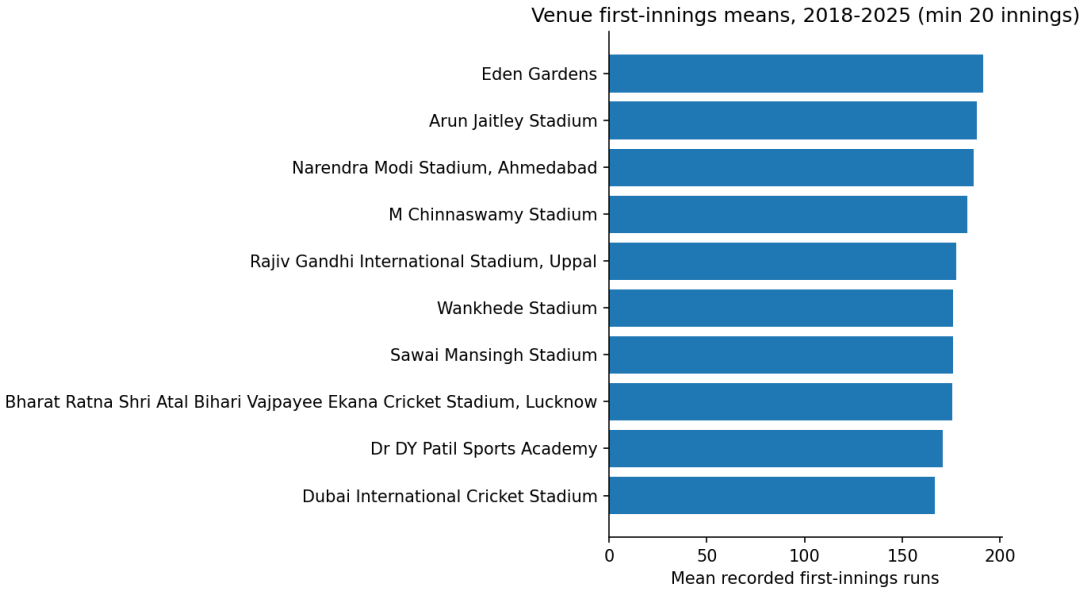
03 Team and ground comparisons

Current franchises: decided-match win percentage, 2022-2025



Gujarat Titans won 37/60 (61.67%) decided games in this interval, compared with Mumbai Indians 26/60 (43.33%). The denominator excludes ties/no-results. These are historical records, not rankings adjusted for schedule strength.

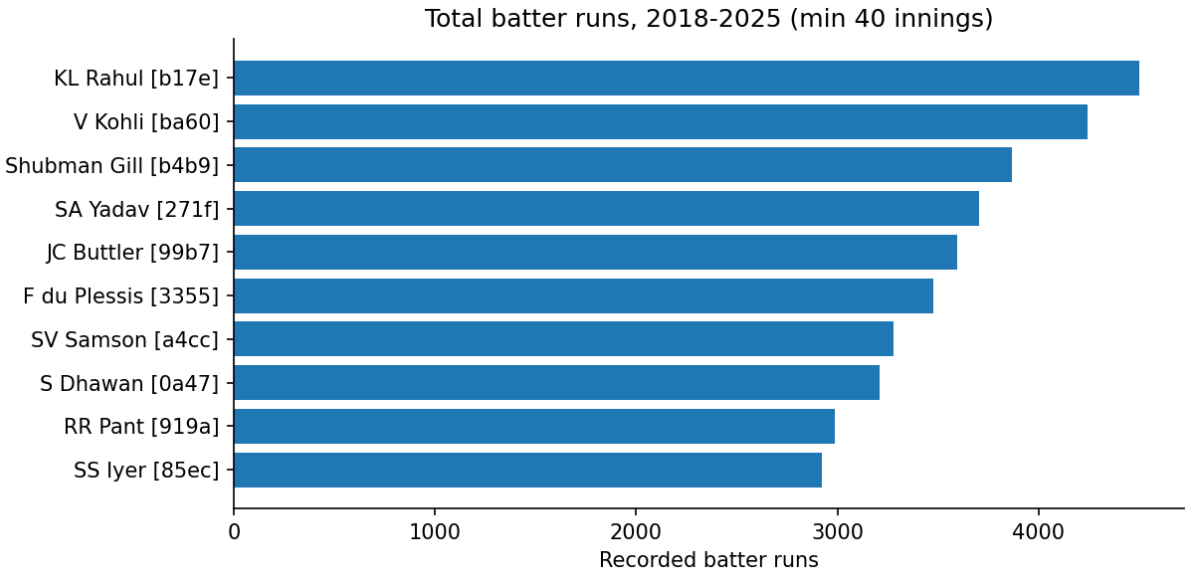
Selected venues: mean first-innings total, 2018-2025



The displayed sample excludes grounds with fewer than 20 first innings. Eden Gardens averaged 191.31 (n=39), while Chepauk averaged 161.47 (n=43). Different teams, seasons and conditions can explain some or all of this difference.

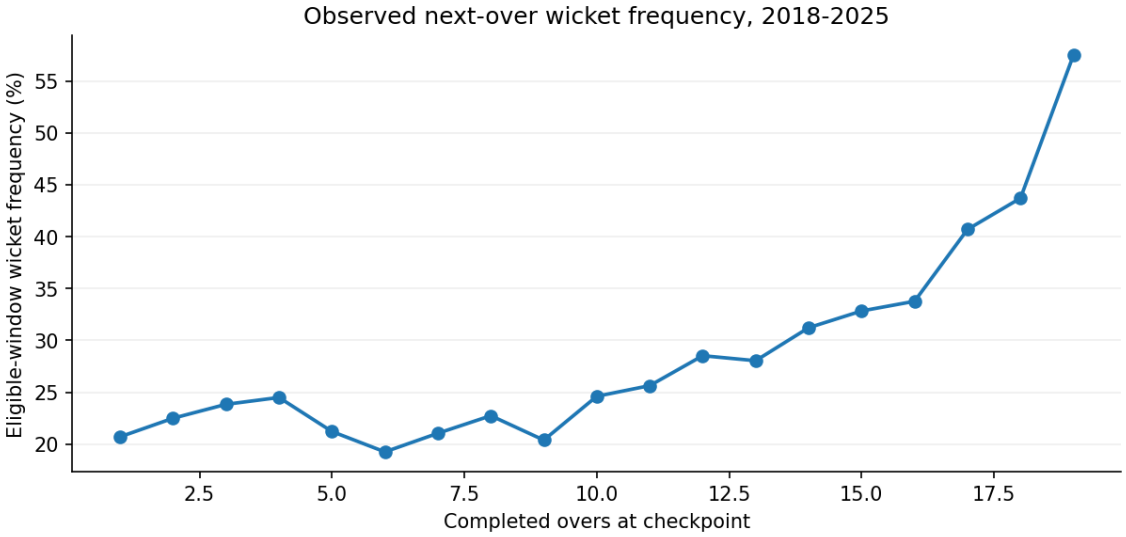
04 Player events and over-by-over outcomes

Highest recorded runs among established batters, 2018-2025



Batters are grouped using stable player IDs; names alone can be ambiguous. Minimum inclusion is 40 recorded appearances. This chart describes observed aggregate runs, not normalized ability or a future-innings prediction.

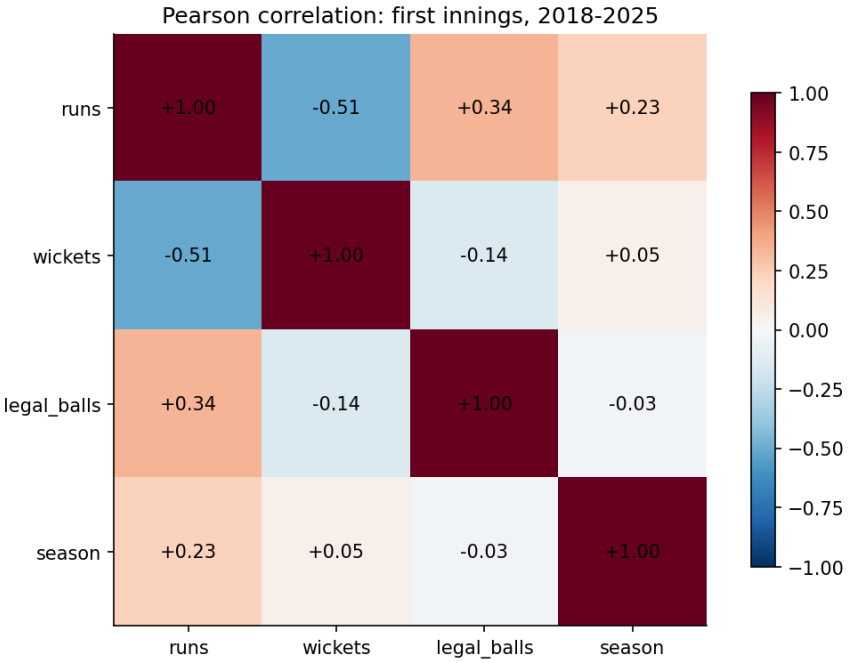
Wicket in the following over: eligible historical checkpoints



Across 19,290 Stage 1 candidate windows in 2018-2025, at least one wicket was recorded in 27.85%. This is a conditional frequency. Censored/shortened windows require further Stage 3 treatment before evaluating an in-match forecasting model.

05 Correlation and descriptive data fitting

Pearson correlation matrix, 2018-2025 first innings



Recorded first-innings runs correlate negatively with final wickets lost (r=-0.507) and positively with recorded legal balls (r=+0.337). Spearman rank correlations are provided in the CSV outputs. Correlation does not establish a causal relationship.

Multiple linear regression - syllabus exercise

Outcome	Predictors	Fit and limitation
Final first-innings runs	Final wickets, legal balls, season	In-sample R^2=0.398; MAE=20.47 runs

Not a third forecasting model. The regressors include values observed only at the END of the innings, so they cannot be used in a pre-match or live score prediction. This is a classical statistical relationship example with in-sample descriptive goodness-of-fit, not evidence of predictive accuracy.

Missingness overview

Among 1,169 pre-2026 match records, 51 city entries are absent (4.36%). Other blank fields reflect ties/no-results or optional outcome methods. The data alone do not prove that missingness is MCAR, MAR or MNAR. The Stage 1 source audit contextualizes city missingness at Dubai and Sharjah.

06 IT462 syllabus alignment and next work

Course topic	Evidence generated in Session 01
Import, data quality, missingness	Stage 1 audited CSVs; missing-value frequencies
Measurement scales and sampling	Runs/wickets numeric-discrete; team/venue nominal; observational archive biases
Central tendency and dispersion	Mean, median, mode, sample SD, sample variance, IQR, MAD, CV, five-number summary
Univariate and bivariate EDA	Histogram, boxplots, team/venue comparisons, phase rates, player scatter
Correlation and regression	Pearson/Spearman matrices, multivariate OLS fit, VIF computations
Advanced EDA / storytelling	Season trend, EDA figures, written caveats and dashboard-relevant data tables

Deliverables and reproduction

Code: src/cricketlens/stage02_eda.py; notebook: notebooks/Stage02_Session01_EDA.ipynb; 11 figures: reports/figures/stage_02/; statistics: reports/stage_02/session_01/; full notes: docs/STAGE_02_SESSION_01_FINDINGS.md.

From the project root, after installing dependencies and placing the seven audited Stage 1 CSVs under data/processed/:

python -m cricketlens.stage02_eda

pytest -q

Next session - Stage 2 is not yet complete

Deepen stratified comparisons and crosstabs; assess confounding by season and conditions; explain the statistical tests and missingness mechanisms; write the consolidated EDA chapter. Only then sign off Stage 2 and start final feature engineering and both ML models in Stage 3.

Academic integrity and references

Source: Cricsheet men's IPL match-by-match JSON (https://cricsheet.org/). Cite the source and comply with its terms. Describe all computational steps accurately and disclose AI assistance as required by IT462. This is a progress report, not a complete evaluated ML or dashboard project.